

**COMCLARK NETWORK & TECHNOLOGY
CORP. / TELTECH**

NATIONWIDE 5G FIXED WIRELESS ACCESS (FWA) INFRASTRUCTURE
PROGRAM

Tender No: RFP-TELTECH-FWA-2026-001
Author: OLIVER TUNGOL
Date: October 6, 2026 · Rev 2.0 (Official
Bidding Version)

REQUEST FOR PROPOSAL (RFP) & TENDER SPECIFICATION

**NATIONWIDE 5G FIXED WIRELESS ACCESS (FWA) NETWORK INFRASTRUCTURE,
SERVICES & MANAGED OPERATIONS**

Tender Document No: RFP-TELTECH-FWA-2026-001

Project Identifier: Nationwide 5G FWA Project (Band n50, 60 MHz TDD)

Issuing Authority: Comclark Network & Technology Corp. / Telecommunications Technology Solutions Inc.
(TelTech)

Author & Lead Architect: OLIVER TUNGOL

Issue Date: October 6, 2026

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Confidentiality: Strictly Confidential — Competitive Commercial Tender

1. EXECUTIVE SUMMARY & INVITATION TO BID

1.1 Purpose of the Tender

Comclark Network & Technology Corp. and Telecommunications Technology Solutions Inc. (hereinafter referred to as the "**Operator**" or "**Company**") hereby invites qualified, world-class telecommunications equipment manufacturers, network systems integrators, and managed service providers (hereinafter referred to as the "**Bidder**" or "**Vendor**") to submit comprehensive, binding technical and commercial proposals for the turnkey supply, delivery, installation, integration, optimization, and managed operations of a nationwide **5G Fixed Wireless Access (FWA) Network**.

The Operator possesses exclusive regulatory spectrum rights for **Band n50 (1432 – 1517 MHz, 1.5 GHz L-Band TDD)** with a **60 MHz contiguous carrier block**, licensed under statutory Republic Act provisions and National Telecommunications Commission (NTC) guidelines.

1.2 Program Scope & Deployment Scale

The nationwide program encompasses a total footprint of **5,000 macro BTS sites** deployed across five (5) consecutive phases:

- **Phase 1 (Day-1 Priority Launch): 1,000 Macro Sites** (Batch 1 — immediate rollout across prioritized unserved and underserved suburban and rural population centers).
- **Phases 2 to 5 (Nationwide Expansion): 1,000 Macro Sites per annum** over Years 2 through 5, reaching full target deployment of 5,000 BTS sites serving over **2.45 to 2.66 Million active broadband subscribers**.
- **Target Addressable Population:** Over **32.3 Million residents** across 5,000 barangays (2024 PSA POPCEN Official Census baseline).
- **Initial Network Sizing Load:** 1,000 active concurrent fixed broadband subscribers per 3-sector BTS (capacity ceiling of 5,965 peak BTS units across expansion).

1.3 Tender Structure & Bid Lots

Bidders are invited to submit tenders for one, multiple, or all of the following distinct Lots:

- **LOT 1: Radio Access Network (RAN) Hardware, Antennas & Turnkey Deployment Services**
- **LOT 2: Power Systems & Passive Site Infrastructure (Co-location Interconnect & Hybrid DC Power)**
- **LOT 3: Cloud-Native 5G Core Network (5GC SA) & Datacenter Infrastructure**
- **LOT 4: Converged Digital OSS/BSS, Field Force Automation & Real-Time Charging Engine**
- **LOT 5: RF Planning, Greenfield Site Survey, Civil Works & Cluster Optimization Services**
- **LOT 6: Nationwide Managed Services, 24/7 Tier-2/Tier-3 NOC & Field Network Operations (MS)**
- **LOT 7: Customer Premises Equipment (CPE) 5G Sub-3GHz High-Gain Terminals**

Bidders submitting integrated "Full Turnkey Consortium" bids spanning Lots 1 through 6 will receive multi-lot synergy evaluation scoring.

2. PROJECT BACKGROUND, OPERATING ENVIRONMENT & TECHNICAL BASELINE

2.1 Spectrum Physics: Band n50 Advantage

Band n50 operates in the 1.5 GHz L-Band (1432 – 1517 MHz), providing superior propagation characteristics over mid-band frequencies (e.g., 3.5 GHz Band n78):

- **Propagation Efficiency:** 1.5 GHz propagation achieves an effective reliable non-line-of-sight (NLOS) and near-line-of-sight (nLOS) coverage radius of **3.5 km** (delivering up to **38.5 km²** of coverage area per site under suburban/rural clutter).
- **Penetration Resilience:** Foliage, atmospheric rain fade, and typical Philippine residential building materials (concrete hollow block, corrugated galvanized sheet roofing) suffer 8–12 dB lower path loss compared to 3.5 GHz.
- **Spectrum Allocation:** 60 MHz contiguous TDD frame structure (configurable TDD DL:UL ratios: 4:1 for broadband downlink-heavy traffic, or 3:1 for balanced enterprise transmission).

2.2 Site Infrastructure Landscape & Tower Strategy

The operator has completed a nationwide spatial overlay matching the 5,000 candidate barangay centroids against **13,746 verified operational cell towers** spanning six (6) tower portfolios:

- **PTCI / MIDC TowerCo:** 5,080 towers

- **FTAP Globe Telecom:** 3,529 towers
- **EdgePoint Infrastructure:** 2,775 towers
- **FTAP Smart Communications:** 1,023 towers
- **LBS Digital Infrastructure (LDIC):** 700 towers
- **FTAP BTS Dedicated Sites:** 639 towers

Deployment Typology (Co-location vs. Greenfield):

- **Class T1 (Immediate Co-location, $\leq 0.5 \text{ km}$):** 54.8% of Phase 1 sites (42.5% nationwide). Sited directly on operational TowerCo structures with existing AC power and civil compound.
- **Class T2 (Close Co-location, $0.5 - 1.0 \text{ km}$):** 26.9% of Phase 1 sites (30.4% nationwide). Sited on operational towers within beamforming reach.
- **Class T3 (Near Co-location, $1.0 - 2.0 \text{ km}$):** 8.0% of Phase 1 sites (17.8% nationwide). Co-location viable subject to RF propagation clearance.
- **Class T4 (Greenfield Monopole Required, $> 2.0 \text{ km}$):** 10.3% of Phase 1 sites (9.3% nationwide; ~460 sites total). Requires self-constructed 30m/40m monopole, civil shelter, dedicated utility transformer drop, and backup diesel/solar generation.

2.3 Transmission & Backhaul Topology

Transmission connectivity has been audited via the independent Codex Optical Backbone screening (1,296 confirmed optical nodes and 4,401 fiber segments):

- **Terrestrial Fiber Optic Cable (FOC):** 98.3% of Phase 1 sites connect via terrestrial fiber drops (Class A: Node $\leq 0.5 \text{ km}$; Class B: Node $\leq 1.0 \text{ km}$; Class C: Line-tap splice closure $\leq 0.5 \text{ km}$; Class D: Lateral/Microwave $\leq 3.0 \text{ km}$).
- **Satellite LEO Backhaul (Starlink Business):** 1.7% of Commercial Phase 1 sites (and up to 34.4%–43.5% in remote universal access/GIDA sites) utilize 4-terminal Starlink Business arrays (delivering aggregated 400–800 Mbps burst with SD-WAN bonding and traffic shaping).

3. SCOPE OF WORK BY TENDER LOT

LOT 1: 5G RAN EQUIPMENT & ACCESS HARDWARE

The Bidder shall supply, deliver, install, test, and commission carrier-grade 3GPP Release 16/17 compliant 5G RAN equipment:

1. Baseband Unit (BBU):

- High-capacity rack-mountable 19" subrack BBU supporting multi-cell aggregation and up to 12 cells of 60 MHz Band n50.
- Dual redundant DC power supply inputs (-48V DC, operating range -40.5V to -57V DC).
- Backhaul interfaces: Minimum 4x 10GE/25GE optical SFP28 interfaces supporting IEEE 1588v2 Precision Time Protocol (PTP) Telecom Profile (G.8275.1) and SyncE.
- Fronthaul interfaces: Minimum 6x 25GE eCPRI interfaces with integrated Optical Supervisory Channel (OSC).

2. Remote Radio Units (RRU) / Active Antenna Units (AAU):

- Configuration: **8T8R MIMO**, 3 sectors per macro site (total of 3 RRUs and 3 antennas per standard macro).
- Operating Frequency: Band n50 (1432 – 1517 MHz), full 60 MHz IBW (Instantaneous Bandwidth) and OBW (Occupied Bandwidth).
- RF Output Power: Minimum **\$8 \times 40 \text{ W}\$ (320 W total RF power per sector).**
- Beamforming capability: Digital horizontal and vertical beamforming supporting dynamic multi-user MIMO (MU-MIMO) pairing.
- Protection & Environment: IP65/IP67 ingress rating, natural convection or high-durability fanless cooling, surviving wind speeds of up to 250 km/h (Typhoon Category 5 resilience).

3. Macro Antennas & RF Ancillaries:

- High-gain 8-port cross-polarized antennas (or integrated AAU) supporting Band n50 with electrical remote downtilt (eRET via AISG 2.0 / 3GPP).
- Horizontal half-power beamwidth: $65^\circ \pm 5^\circ$; Minimum gain $\geq 17.5 \text{ dBi}$.
- Low PIM performance: $\leq -153 \text{ dBc}$ @ $2 \times 43 \text{ dBm}$.
- High-grade low-loss optical fiber patchcords, IP67 weatherproof boot assemblies, DC power surge protective devices (SPD Type 1+2), and grounding kits.

LOT 2: POWER SYSTEMS & PASSIVE INFRASTRUCTURE

The Bidder shall provide unified site power solutions engineered for high availability across both leased TowerCo sites and greenfield sites:

1. DC Power Rectifier Plant:

- Modular 19" 3U/4U switchmode DC power system dimensioned for **6.0 kW** continuous capacity ($4 \times 2.0 \text{ kW}$ or $3 \times 3.0 \text{ kW}$) hot-swappable rectifier modules in $N+1$ redundancy).
- Rectifier efficiency: $\geq 96.0\%$ (80 PLUS Titanium / high-efficiency class).
- Advanced SMPS controller with SNMPv3, HTTPS, Modbus, dry contacts, and remote cloud telemetry.

2. Battery Energy Storage System (BESS):

- Lithium Iron Phosphate (LiFePO₄ / LFP) battery modules: Minimum **$1 \times 48 \text{ V} / 150 \text{ Ah}$ (or $2 \times 100 \text{ Ah}$)** providing $\geq 4 \text{ hours}$ backup autonomy at standard 2.5–3.0 kW BTS operational load.
- Cycle life: $\geq 4,000$ cycles at 80% Depth of Discharge (DoD) @ 25°C .
- Built-in intelligent BMS with cell-level balancing, over-charge, over-discharge, short-circuit, and high-temperature thermal runaway cutoffs.

3. Standby Generator Sets (Greenfield Class T4 Sites):

- 15 kVA / 12 kW prime-rated, 230V single-phase / 400V 3-phase, 60 Hz water-cooled diesel generator with sound-attenuated weatherproof enclosure ($< 68 \text{ dBA}$ @ 7m).
- Minimum 500-liter to 1,000-liter sub-base day fuel tank providing $\geq 72 \text{ hours}$ continuous operation during grid blackout.
- Automatic Transfer Switch (ATS) with motorized 4-pole contactor and mechanical interlock.

4. TowerCo Co-Location Power Interconnect Kit:

- Standardized DC tap-off distribution box, sub-metering Class 1.0 revenue-grade energy meter with RS485 telemetry, and dedicated surge suppression.

LOT 3: CLOUD-NATIVE 5G CORE (5GC SA) & DATACENTER INFRASTRUCTURE

The Bidder shall deploy a carrier-grade, cloud-native 5G Standalone (5G SA) Core Network dimensioned for **1,000,000 initial registered subscribers** scaling up to **3,000,000 subscribers**, deployed in a **Geo-Redundant, Active-Active Dual Datacenter Topology** (NCR Primary DC and Metro Clark / Region III Disaster Recovery DC):

1. Network Functions (NFs) Scope:

- Control Plane: AMF, SMF, NRF, NSSF, PCF, UDM, UDR, AUSF, NEF, CHF.
- User Plane: High-performance distributed User Plane Function (UPF) nodes.

2. User Plane (UPF) Throughput Dimensioning:

- UPF capacity dimensioned for **80 Gbps usable throughput per 2U dual-socket compute node** (minimum \$2 \times 100 \text{ GE} line-rate DPDK / SR-IOV interfaces).
- Integrated high-performance **Carrier-Grade NAT (CG-NAT)** supporting deterministic port allocation and \$10:1 to \$16:1 subscriber port density.
- Integrated Deep Packet Inspection (DPI) for layer-7 traffic classification, fair-use bandwidth throttling, and URL filtering.

3. Hardware & CaaS Virtualization Infrastructure:

- Enterprise OpenStack or Bare-Metal Kubernetes (CaaS) orchestration platform.
- Dual-socket Intel Xeon / AMD EPYC server nodes, 512 GB RAM, dual 25G/100G NICs, redundant enterprise NVMe storage arrays.
- Redundant datacenter leaf-spine fabric switches (\$32 \times 100 \text{ GE} spine, \$48 \times 25 \text{ GE} + 6 \times 100 \text{ GE} leaf), redundant perimeter next-gen firewalls, and DDoS mitigation scrubbing appliances.

4. Licensing Framework:

- Tiered perpetual subscriber capacity licensing:
- Tier 1 (First 100k subs): Bidder to quote per registered sub.
- Tier 2 (100k to 500k subs): Bidder to quote per registered sub.
- Tier 3 (500k to 1M+ subs): Bidder to quote per registered sub.

LOT 4: CONVERGED DIGITAL OSS/BSS & FIELD FORCE AUTOMATION

The Bidder shall deliver an integrated, cloud-native Telecom Business Support System (BSS) and Operations Support System (OSS) optimized for high-volume, low-friction fixed broadband operations:

1. Converged BSS & Omnichannel Customer Engagement:

- Omnichannel CRM, Product Catalog, and CPQ supporting 4 primary retail tariffs and SME tiers.
- Native Prepaid-First architecture: Integration directly with 5G Core Charging Function (CHF) via 3GPP Nchf / Diameter Ro interfaces.
- Payment Gateway Integrations: Native, production-ready REST API integrations with Philippine digital payment channels: **GCash, Maya (PayMaya), 7-Eleven CLiQQ OTC, Bayad Center**, bank transfer (InstaPay/PESONet), and municipal distributor airtime portals.

2. Field Force Management (FMS) & Mobile Technician App:

- Automated work-order generation and algorithmic dispatch engine handling up to **3,500 daily work orders** (new installs, truck rolls, relocations, CPE reclaims, and trouble calls).
- Android mobile app for field technicians: Real-time GPS dispatch, barcode/QR scanning of CPE MAC and serial numbers, automated outdoor signal sweep tool (RSRP/SINR validation), photograph proof of installation, and customer digital e-signature.
- Turnaround SLA: Compressed installation timeline from 14 calendar days down to **under 48 hours** from order booking.

3. CPE Warehouse Logistics & Asset Reverse Logistics (WMS):

- End-to-end serialized asset tracking from central warehouse staging through regional depots, technician truck float, subscriber premises, and repossession/refurbishment cycle.
- Support for minimum 70% CPE recovery and refurbishment upon subscriber churn.

4. OSS Network Management, Fault & Performance Monitoring:

- Multi-vendor RAN EMS/NMS integration via 3GPP northbound REST/Kafka/SNMP interfaces.
- Real-time automated fault correlation, root-cause analysis (RCA), and synthetic SLA monitoring.
- Automated NTC regulatory compliance KPI dashboard and QoS reporting engine.

LOT 5: RF PLANNING, SITE SURVEY & CLUSTER OPTIMIZATION

The Bidder shall execute end-to-end RF planning, site survey, civil design, and cluster radio tuning:

1. Detailed Greenfield & Co-Location Site Surveys:

- Technical Site Survey Report (TSSR) for each assigned candidate barangay: Detailed structural load analysis on existing towers, antenna mounting space, azimuthal line-of-sight (LOS) drone photographic sweeps (360° at 20m , 30m , 40m AGL), AC power distribution panel capacity, grounding resistance ($\leq 5\text{ Ohms}$), and backhaul cable routing.

2. High-Precision RF Propagation Modeling:

- Production-grade RF propagation model tuning in Asset / Atoll / Planet utilizing high-resolution GIS clutter and 3D digital elevation models (DEM $\leq 5\text{m}$ resolution).
- Cell dimensioning: Sizing for 60 MHz Band n50 TDD, 3 sectors @ 120° nominal separation, antenna height 30m–45m AGL, coverage target RSRP $\geq -95\text{ dBm}$ across $\geq 90\%$ of populated barangay settlement areas.

3. Single Site Verification (SSV) & Cluster Tuning:

- SSV drive tests and stationary multi-point testing per site verifying 3-sector RF radiation, antenna tilt (RET), PCI validation, azimuth verification, and handover execution.
- Cluster optimization across 10–20 site clusters to eliminate pilot pollution, optimize neighbor lists, minimize inter-cell interference, and balance sector traffic loading.

LOT 6: NATIONWIDE MANAGED SERVICES & 24/7 NETWORK OPERATIONS (MS)

The Bidder shall provide fully managed, SLA-governed network operations, field maintenance, and spare parts management:

1. Tier-2 & Tier-3 Network Operations Center (NOC):

- 24/7/365 active monitoring of RAN, Core, Transmission, Power, and OSS/BSS platforms.
- Incident management, event correlation, dispatching field maintenance teams, and managing customer trouble ticket escalations.

2. Nationwide First-Line Maintenance (FLM) Field Operations:

- Regional field maintenance hubs across Greater Luzon, Northern Luzon, Southern Luzon, Visayas, and Mindanao.
- Mean Time to Respond (MTTR) ≤ 2 hours in urban/suburban and ≤ 4 hours in rural clusters for Critical/Severity-1 incidents.
- Preventive maintenance: Quarterly inspection of RF antenna brackets, coax/optical jumpers, rectifier modules, battery conductance testing, and generator load testing.

3. Comprehensive Spare Parts Management (SPMS):

- Advanced replacement logistics: Central warehouse and regional staging depots maintaining an audited safety float of BBUs, RRUs, rectifiers, lithium batteries, and SFPs.
- Guarantee of next-business-day (NBD) on-site replacement delivery for failed hardware components.

LOT 7: 5G HIGH-GAIN CUSTOMER PREMISES EQUIPMENT (CPE)

The Bidder shall supply outdoor and high-gain indoor 5G FWA CPE devices:

1. RF & Modem Specifications:

- Chipset: 3GPP Release 16 compliant 5G NR sub-3GHz processor.
- Frequency Bands: Native Band n50 support (1432 – 1517 MHz, 60 MHz channel bandwidth) plus fallback sub-6GHz bands (n1, n3, n7, n28, n41, n77, n78).
- Antenna: Internal high-gain directional/omnidirectional antenna array (≥ 8 to 10 dBi) antenna gain in Band n50; optional external TS-9/SMA antenna ports.
- MIMO: 4R MIMO receiver architecture in Band n50 downlink.

2. LAN & Local Wi-Fi Specifications:

- Wi-Fi 6 (802.11ax), dual-band 2.4 GHz + 5.0 GHz, 2×2 MU-MIMO, aggregate speed ≥ 1800 Mbps (AX1800).
- Ethernet: Minimum 1×2.5 GE + 2×1 GE RJ-45 LAN ports.
- Voice: $1 \times$ FXS RJ-11 VoIP port (optional).

3. Device Management & TR-069 / USP (TR-369):

- Full remote provisioning via TR-069 / TR-181 or TR-369 USP; remote firmware OTA (FOTA) upgrades; SIM-lock and network-lock enforcement tied to Operator PLMN (515-XX).

4. RIGOROUS SERVICE LEVEL AGREEMENTS (SLAs) & PERFORMANCE KPIS

The Bidder must guarantee compliance with the following international telecom best-practice KPIS. Failure to achieve these targets shall trigger contractual liquidated damages (LDs) and service credits.

4.1 Radio Access Network (RAN) Performance KPIs

KPI Metric	Formula / Standard	Minimum Acceptance	Target Excellence	Measurement Window
Radio Network Availability	$\frac{\text{Total Available Cell Hours} - \text{Cell Outage Hours}}{\text{Total Available Cell Hours}} \times 100\%$	$\geq 99.70\%$	$\geq 99.95\%$	Monthly / 24x7
5G Radio Accessibility (CSSR)	Call/Session Setup Success Rate (RRC + 5G QoS Flow)	$\geq 99.00\%$	$\geq 99.60\%$	Busy Hour
5G Session Drop Rate (CDDR)	Abnormal Release Rate per 100 established sessions	$\leq 0.50\%$	$\leq 0.15\%$	Busy Hour
Single Sector User Capacity	Active concurrent connected CPE sessions per sector	$\geq 350 \text{ CPEs}$	$\geq 450 \text{ CPEs}$	Peak Busy Hour
Peak Cell DL Throughput (60 MHz)	8T8R 60 MHz TDD peak aggregated sector throughput	$\geq 450 \text{ Mbps}$	$\geq 520 \text{ Mbps}$	Lab & Single Cell Test
Cell Edge DL User Throughput	95th percentile cell-edge subscriber download rate	$\geq 25 \text{ Mbps}$	$\geq 50 \text{ Mbps}$	Busy Hour
Cell Edge UL User Throughput	95th percentile cell-edge subscriber upload rate	$\geq 5 \text{ Mbps}$	$\geq 15 \text{ Mbps}$	Busy Hour
End-to-End User Latency (RTT)	CPE to UPF packet round-trip time	$\leq 30 \text{ ms}$	$\leq 18 \text{ ms}$	Continuous Ping
DL Packet Loss Rate (PLR)	User Plane packet discard rate	$\leq 0.10\%$	$\leq 0.01\%$	Monthly Average

4.2 Power & Site Infrastructure Resilience KPIs

Infrastructure Metric	Technical Requirement	Minimum SLA Target	Failure Remedy
Site Power Availability	BTS Continuous Operating Power (AC Grid + DC Battery + Genset)	$\geq 99.95\%$ (Leased) $\geq 99.85\%$ (Greenfield)	Service credit deduction per hour outage
Battery Autonomy Duration	Time BTS operates on LFP battery during AC utility blackouts	$\geq 4.0 \text{ Hours}$ @ 3.0 kW load	Mandatory module replacement if $< 3.5 \text{ hrs}$
Generator Auto-Start Reliability	Genset starts and ATS transfers load within 60s of grid failure	$\geq 99.0\%$ successful transfers	Liquidated damages on site drop
Grounding Earthing Resistance	Earth ground electrode resistance to true earth	$\leq 5.0 \text{ Ohms}$	Mandatory soil conditioning / additional rods
Rectifier System Efficiency	Operating efficiency from 30% to 90% load range	$\geq 96.0\%$	Rejection of non-compliant modules

4.3 5G Core Network & Cloud Infrastructure KPIs

Core Metric	Operational Definition	Target SLA Benchmark
5GC Control Plane Availability	High availability of AMF, SMF, UDM, PCF across geo-cluster	$\geq 99.999\%$ ("Five Nines")
5GC User Plane Availability	Availability of UPF packet routing instances	$\geq 99.999\%$
PDU Session Establishment Success	Initial PDU session setup success rate	$\geq 99.80\%$
Session Setup Delay	End-to-end signaling time to establish 5G session	$\leq 80\text{ ms}$ (95th percentile)
UPF Packet Forwarding Latency	Processing delay through UPF hardware	$\leq 1.0\text{ ms}$
CG-NAT Port Allocation Success	Real-time port assignment for new outbound sessions	$\geq 99.99\%$
Core Failover Recovery Time (DR)	Automatic cutover from NCR DC to Clark DR DC	$\leq 30\text{ seconds}$ (Zero session loss)

4.4 OSS/BSS, Field Operations & Managed Services SLAs

Operational Service Domain	Metric Description	Mandatory SLA Benchmark
BSS Billing & Rating System Availability	Online Charging System (CHF/OCS) uptime	$\geq 99.99\%$
Payment Top-Up Processing Time	Time from payment channel confirmation to account credit	$\leq 3\text{ seconds}$
Field Force Installation Lead Time	Order placement to physical CPE on-site activation	$\leq 48\text{ Hours}$ (Standard urban/suburban)
First-Time Right (FTR) Installs	Installations operating without trouble call within 14 days	$\geq 96.0\%$
Customer Churn CPE Recovery Rate	Successful physical recovery of CPE on churned accounts	$\geq 70.0\%$
Severity 1 Incident MTTR (Critical Outage)	Time to restore full service on macro BTS outage	$\leq 2\text{ Hours}$ (Urban) / $\leq 4\text{ Hours}$ (Rural)
Severity 2 Incident MTTR (Major Degraded)	Time to restore single sector failure or degraded backhaul	$\leq 6\text{ Hours}$
Hardware Replacement Turnaround	Depot dispatch of spare BBU/RRU/Rectifier to site	$\leq 24\text{ Hours}$ (Nationwide NBD)

5. TENDER SUBMISSION STRUCTURE & COMPLIANCE REQUIREMENTS

Bidders must submit their proposals divided strictly into three (3) separate volumes:

Volume I: Executive & Commercial Proposal (Financial Envelope)

1. **Form A — Bid Submission Letter & Consortium Structure:** Executed by authorized corporate officers.
2. **Form B — Bill of Quantities (BoQ) & Unit Price Book:** Exhaustive pricing matrix in USD and PHP for all hardware, software licenses, implementation services, and annual maintenance contracts (AMC), completed using the official blank Excel BoQ template.
3. **Form C — Total Cost of Ownership (TCO) Breakdown (10-Year Horizon):** Itemized Day-1 CAPEX, expansion CAPEX, and 10-year OPEX schedules.
4. **Form D — Vendor Financing Proposal (Mandatory Requirement):**
 - The Bidder **must** provide a detailed vendor financing facility covering at least **80% of Capex** (BTS hardware, core, and turnkey rollout).
 - Tenor: Minimum 4-year to 5-year repayment facility.
 - Interest Rate: Fixed commercial rate benchmarked to SOFR + spread.
 - Grace Period: Minimum 12-month grace period on principal repayment during Phase 1 deployment.

Volume II: Technical Architecture & Compliance Matrix (Technical Envelope)

1. **Section A — RAN Technical Design:** 3GPP R16/17 compliance, Band n50 8T8R AAU/RRU datasheets, BBU subrack capacity, antenna radiation patterns, and link budget calculations.
2. **Section B — Power & Site Civil Engineering:** Rectifier plant, LFP battery cycle curves, genset specifications, and TowerCo co-location interface diagrams.
3. **Section C — 5G Core Architecture:** Cloud-native microservices architecture, dual datacenter active-active geo-redundancy topology, UPF DPDK hardware dimensioning, and CG-NAT sizing.
4. **Section D — OSS/BSS & IT Stack:** Modular architecture, digital self-care portal, payment channel API integrations, field-force management mobile application, and WMS inventory tracking.
5. **Section E — RF Planning & Cluster Optimization Methodology:** Link budget models, clutter definitions, drive-test methodology, and SSV acceptance criteria.
6. **Section F — Managed Services & Operations Plan:** NOC structure, FLM staffing model, SLA penalty framework, and spare parts dimensioning model.
7. **Compliance Matrix:** Line-by-line statement of compliance (**Compliant / Minor Deviation / Non-Compliant**) against every technical requirement in Appendix A.

Volume III: Bidder Qualifications, References & Financial Soundness

1. **Corporate Financial Statements:** Audited financial statements for the past three (3) fiscal years demonstrating solvency, liquidity, and operational turnover.
2. **Telecom Track Record:** Proof of at least three (3) commercial 5G deployments (RAN or Core) of similar or larger scale within the past 36 months.
3. **Philippine In-Country Support Structure:** Documented local Philippine engineering entity, SEC registration, local repair center, and existing field logistics capabilities.

6. BID EVALUATION CRITERIA & AWARD METHODOLOGY

Tenders will be evaluated using a **Quality and Cost-Based Selection (QCBS)** scoring framework:

$$\text{Total Evaluation Score} = (\text{Technical Score} \times 60\%) + (\text{Commercial \& Financing Score} \times 40\%)$$

6.1 Technical Evaluation Breakdown (100 Points / 60% Total Weight)

- **RAN Architecture & Radio RF Performance (25 Points):** Band n50 output power, 8T8R beamforming algorithms, power consumption efficiency, and environmental durability.
- **5G Core Architecture & Geo-Redundancy (20 Points):** Cloud-native architecture, UPF line-rate throughput, CG-NAT scale, and seamless hitless failover.
- **OSS/BSS Capabilities & Local Integrations (15 Points):** Real-time charging, native Philippine digital payment interfaces, automated field dispatch tool, and WMS inventory reverse logistics.
- **Power Engineering & Site Solution (10 Points):** LFP battery cycle life, rectifier efficiency, TowerCo co-location integration simplicity, and genset reliability.
- **RF Planning, Optimization & Rollout Methodology (15 Points):** Link budget precision, cluster tuning rigor, and proven ability to mobilize 1,000 sites in 12 months.
- **Bidder Experience, References & Local Entity (15 Points):** Tier-1 telecom reference base, local Philippine support team, and spare parts warehouse infrastructure.

6.2 Commercial & Financing Evaluation Breakdown (100 Points / 40% Total Weight)

- **Capex Competitiveness (40 Points):** Evaluated across lowest compliant Capex bids. Lowest qualifying responsive Capex receives full points.
 - **10-Year Opex & TCO Efficiency (25 Points):** Low ongoing power consumption, competitive software AMC rates, and managed services unit pricing.
 - **Vendor Financing Structure (35 Points):** Grace period length, interest rate spread, financing quantum (\$\ge 80\%\$), and ease of financial covenants.
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7. TENDER TIMELINE & KEY MILESTONES

Milestone Event	Scheduled Date	Responsibility
Official RFP Tender Issuance	October 6, 2026	Operator Tender Committee
Bidder Registration & NDA Signing	October 16, 2026	Prospective Bidders
Pre-Bid Clarification Meeting (Hybrid)	October 23, 2026	Operator & Bidders
Deadline for Written Inquiries / RFIs	October 30, 2026	Prospective Bidders
Operator Formal Addendum & Clarifications	November 6, 2026	Operator Tender Committee
Tender Submission Deadline (Sealed Bids)	November 27, 2026 (17:00 PHT)	Bidders
Opening of Technical Envelopes	November 30, 2026	Operator Technical Board
Technical Clarifications & Vendor Presentations	December 1 – 11, 2026	Shortlisted Bidders
Opening of Commercial & Financing Envelopes	December 15, 2026	Operator Commercial Board
Final Contract Negotiation & Award	January 15, 2027	Operator & Winning Bidder(s)
Phase 1 Contract Execution & Kickoff	February 1, 2027	Operator & Appointed Contractor
Phase 1 Target Commercial Launch (1,000 Sites)	Q4 2027 (Within 12 Months)	Joint Deployment Team

8. INSTRUCTIONS TO BIDDERS & SUBMISSION DETAILS

8.1 Clarifications and Communications

All formal communications, requests for interpretation, and tender queries must be submitted in writing via official email to:

- **Tender Secretariat:** fwa.tender@comclark.com.ph
- **Attention:** Technical Evaluation Board / OLIVER TUNGOL (Author & Lead Architect)
- **Subject Line:** [RFP-TELTECH-FWA-2026-001] Technical Clarification Request –

8.2 Submission Format

- **Electronic Submission:** One (1) encrypted USB storage device containing PDF and editable Excel files of all volumes, plus cloud upload to the secure Operator Tender Portal.
- **Physical Submission:** Two (2) complete hard-copy sets (One Original, One Duplicate) of Volume I, Volume II, and Volume III, sealed in wax-stamped envelopes delivered to:
- *Tender Committee, Comclark Network & Technology Corp. / TelTech Corporate Headquarters, Metro Manila, Philippines.*

(End of Request for Proposal Specification — Tender Appendices Follow)